Reference-Free Failure Review for Camera Tracking: Protocol and Preliminary Experiments

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Abstract

Camera trajectories are often estimated on videos without reference camera poses. We study how to assess whether such estimates are usable, while separating malformed outputs, functional failures, and insufficient evidence. Our proposed protocol combines source images, independently selected image correspondences, deterministic geometric measurements, and separately validated visual interpretation. Every attempted sequence remains in the accounting, including failures and abstentions.

This report presents a development implementation, not a validated metric. On three 60-frame DAVIS training clips, we retain DAGE and Depth Anything 3 estimates and compare geometric consistency and early-fit, later-frame point placement. Translation-sign controls expose an ambiguity of epipolar residuals. Image-track availability sharply limits placement comparisons: on the BMX clip, replacing Lucas–Kanade with CoTracker3 increases common support from 3 to 98 of 588 planned observations. Separate prefix inference prevents future images from changing the coordinates used for the early point fit. Independent usability judgments, comparison with existing reference-free metrics, and held-out validation remain incomplete. The intended contribution is a validated failure-review methodology; the present results do not yet establish it.

1 Introduction

A camera trajectory can have a good global alignment score and still look wrong. It may shake from frame to frame, rotate erratically, pause while the video moves, or fail for a short interval. Absolute trajectory error (ATE), relative pose error (RPE), and pose-threshold accuracy each reveal part of the result. None is a complete account of visible quality.

This creates a practical problem when choosing between camera-tracking methods. Researchers inspect videos, plots, point clouds, and per-frame errors, then make an informal judgment. The judgment may be useful, but it is hard to repeat. The views can differ between methods, labels may reveal the method, invalid outputs may disappear, and a final ranking may have no uncertainty estimate.

We propose Plumbline Review, a reproducible evaluation layer for camera tracking. It does not replace pose metrics. Every candidate first passes deterministic validity checks. Geometric and image evidence then support localized tests of the target task, while unavailable evidence is recorded explicitly. Our current placement test fits a static point from early image observations and projects it into later frames without refitting. Source-video overlays show the observed image feature and the predicted marker together. Such a discrepancy is diagnostic, not sufficient to assign camera failure: image tracks, calibration, and point fitting can also be wrong. Blinded pairwise review remains a supporting protocol rather than the main contribution.

This design draws on pairwise evaluation in 3DCodeBench, where automated measures alone do not capture all differences in generated 3D assets [4]. The camera problem requires added controls: trajectory conventions, Sim(3) alignment, non-finite poses, frozen tracks, and separation between automatic rejection and subjective review.

This development report contains:

1. an implemented separation of structural failure, functional judgment, and unresolved evidence;
2. reference-free source and geometric diagnostics with retained input and output provenance;
3. controlled sensitivity measurements, placement-evidence coverage tests, and held-out validation of a correspondence check; and
4. a validation protocol for the proposed evaluator, with pairwise review and design search retained as supporting components.

2 Related Work

Evaluation without reference poses. Object Reprojection Error (ORE) evaluates camera trajectories using static-object tracklets and camera intrinsics rather than

reference camera poses [2]. Its EgoStatic benchmark applies this approach to existing egocentric videos. Thus, evaluation without pose ground truth is not itself our contribution. Our proposed extension concerns human-validated functional failure judgments with localized evidence and abstention. Demonstrating value beyond ORE remains necessary; the current experiments do not establish that result.

Camera metrics. Trajectory benchmarks commonly use global alignment followed by ATE, local relative errors, or pairwise rotation and translation thresholds. Recent geometric models report several of these measures, including DUSt3R [13], MASt3R [7], MonST3R [14], CUT3R [12], VGGT [11], and Depth Anything 3 [8]. Multiple metrics help, but researchers still inspect outputs when the numbers disagree.

Reproducible evaluation. Clear protocols, reusable code, and retained artifacts improve reproducibility [10]. Evaluation harnesses in language modeling have shown that small implementation choices can alter conclusions [1]. Our work applies the same principle to camera review: the render, gate, pairing, rubric, and aggregation rule are part of the measurement.

Preference evaluation. 3DCodeBench combines automated measures with pairwise human preference for procedural 3D models [4]. Our supporting pairwise protocol applies this approach to camera diagnostics; its connection to design search is proposed, not experimentally established.

Learned visual evaluators. DB-3DME adapts a VLM's visual encoder using human ratings of generated meshes [5]. 3D-DefectBench studies how model choice, rendering, and evidence affect defect judgments [15]. These works preclude treating VLM judging or evaluator fine-tuning alone as novel. Our intended distinction is validated camera-specific functional failure assessment, with geometric checks and abstention; that distinction remains to be demonstrated experimentally.

3 Method

3.1 Functional failure and unresolved evidence

The target application is recovering camera motion suitable for stable placement of static 3D content. A process crash, timeout, or malformed pose array is a mechanical failure. A structurally valid estimate can still be unusable for that task; we call that a functional DNF. A plausible-looking trajectory is not sufficient evidence of usability. Missing calibration, ambiguous motion, or uninformative correspondences can instead require abstention. The current implementation records these categories, but no automated functional-DNF rule has been validated.

Let N be the number of attempted sequences, M the mechanical failures, F the reviewed functional failures, U the reviewed usable results, and A the unresolved results, so $N = M + F + U + A$. We report all five counts and the failure and unresolved fractions. The interval $[(M + F)/N, (M + F + A)/N]$ describes unresolved outcome accounting, not a statistical confidence interval. Metrics conditioned on usable outputs must be accompanied by their sample counts and the all-attempt failure accounting. Failed outputs are never silently removed from the denominator.

3.2 Reference-free evidence and proposed interpretation

The implemented DAVIS diagnostics combine sampled source frames with the predicted path, orientation glyphs, and between-frame motion. They do not use reference poses or invent capture timestamps. Independent SIFT matching supplies geometric evidence: matches are selected without candidate poses, then tested against those poses and the model's estimated intrinsics. We retain every selected match, including possible false matches and moving objects. A low epipolar residual alone cannot establish a correct trajectory, and pure rotation or negligible translation can make this evidence uninformative.

The proposed learned component must interpret scene evidence without replacing numerical measurements. Potential roles include identifying regions whose motion is independent of the camera and recognizing when evidence is insufficient. These roles are not yet implemented and validated as a complete evaluator. Point–line measurements are computed deterministically. Model explanations must not contradict those measurements. Training an evaluator will require independently checked labels; neither perturbation magnitude nor an unvalidated model answer supplies a usability label.

3.3 Early-fit placement evidence and coverage

We divide the first source image into a fixed 4×3 grid and select the strongest eligible corner in each cell, using an eroded DAVIS background mask. Empty cells remain in the accounting. Selection does not use a candidate's poses or projection errors. Background annotation is not a static-scene certificate.

For each candidate, a 3D point is fit to the rays from frames 0–10 using that candidate's poses and intrinsics. The fitted point remains fixed when projected into frames 11–59. A degenerate fit or a point behind a fitting camera is unavailable, not a zero-error placement. For each later observation we retain projection error, image-track availability, visibility checks, and annotation membership. Comparisons use the same observations

for both candidates and report coverage out of the full $12 \times 49 = 588$ planned observations per clip. This is a joint pose/calibration/point-fit consistency diagnostic, not pose error in isolation.

The Lucas–Kanade (LK) arm uses a 1-pixel forward/backward check and does not restart a lost track. An additional direct first-frame-to-current-frame pass provides a sensitivity check, with a 3-pixel agreement criterion; it is not independent verification. The CoTracker3 arm [6] uses the same query locations and retains the upstream predictor's visibility and support-grid behavior. Its offline inference is run separately on the early prefix and the full video. Only prefix-only coordinates enter the point fit. Otherwise, later images can influence early coordinates even when the geometric fitting code uses only early frame indices. None of these development thresholds is a validated correctness or functional-DNF threshold.

3.4 Review artifact

For system i on task t, the harness writes an immutable artifact

$$A_{it} = \{M, P, D, G, V\},$$

where M is a manifest, P is pose data, D is diagnostic evidence, G contains available scalar measurements, and V records schema, renderer, code, and configuration versions. Reference poses are optional and are explicitly absent in the DAVIS arm. Dataset, sequence, system, and configuration identifiers are stored in the manifest. File hashes make later changes visible.

The existing reference-assisted diagnostic has fixed size and layout. It shows aligned paths from consistent viewpoints, per-frame normalized translation error, rotation error, and a metric panel. Pair images place two diagnostics side by side and use only labels A and B. A deterministic hash changes left-right order, so an interrupted run can resume without changing the assignment. This diagnostic is separate from the reference-free source views; results from the two protocols must not be pooled as the same review task.

3.5 Validity before preference

Preference is not used to rescue malformed output. The structural gate checks pose-array shape, frame count, finite coverage, homogeneous rows, rotation orthogonality, and rotation determinant. Frozen motion is not a structural failure: a stationary camera can legitimately produce a constant trajectory. Functional usability requires a separate judgment. Invalid reference evidence blocks the comparison rather than generating candidate-failure labels. With valid reference evidence, a pair containing one structurally invalid candidate receives an automatic outcome; two structurally invalid candidates are recorded as *both bad*. Degenerate alignment is recorded as unavailable diagnostic evidence, not as a functional failure or a perfect score.

Valid trajectories are aligned to the reference with one global Sim(3) transform for diagnostic position errors. Rotation errors are computed separately. Alignment removes coordinate gauge but does not remove temporal jitter or a local failure.

3.6 Supporting reference-assisted pairwise review

For each task, the scheduler chooses the least-observed system pairs. Ties in global pair count are resolved lexically, which makes scheduling reproducible. The reviewer sees the pair image and applies a fixed rubric:

1. Does either trajectory contain a local jump or frozen interval?
2. Which trajectory has less frame-to-frame jitter?
3. Which rotation track is more stable and consistent with the reference?
4. Which trajectory better preserves the path over the full sequence?

The response is A, B, tie, or both bad. The log stores the comparison identifier, reviewer class, verdict, reason, artifact hashes, and time. System identities are joined only during aggregation.

An external vision-language judge can receive the same image and response schema. It is not treated as interchangeable with a person. Before use, its non-tie coverage, accuracy on shared decisive comparisons, confusion matrix, and Cohen's κ are measured against held-out human votes. The present paper tests the provider-neutral interface but does not report a human-calibrated model judge.

3.7 Preference model and uncertainty

For a decisive comparison between systems i and j, the Bradley–Terry model is

$$P(i \succ j) = \frac{\exp(q_i)}{\exp(q_i) + \exp(q_j)}.$$

A tie contributes one half-win to each system. Both-bad pairs do not enter the fit. A small L_2 term prevents numerical separation, and scores are centered because only differences are identifiable. The comparison graph must be connected.

Confidence intervals resample complete tasks rather than individual votes. This keeps all comparisons from one video together and avoids treating correlated views of the same scene as independent evidence. We report coverage and invalid-rate results next to preference scores.

3.8 Supporting design-search implementation

A validated evaluator could provide an expensive black-box objective for configuration search. We have implemented a proposer, but have not demonstrated this connection on real tracking quality. Parameters may be continuous, integer, or categorical. Completed observations contain the configuration, preference estimate or controlled quality score, runtime objectives, constraints, and status.

With fewer than three successful observations, the proposer uses maximin space filling. It then fits Gaussian-process surrogates with an RBF kernel. Multi-objective runs use a seeded random scalarization, following the ParEGO pattern, rather than one hidden permanent weighting. Expected improvement chooses the next candidate. Separate surrogates downweight candidates likely to violate constraints or fail. Every proposal stores its seed, objective weights, acquisition value, and configuration hash.

This loop does not make the test set part of training. Search uses a declared development set and budget; the chosen configuration is evaluated once on a held-out final set.

4 Controlled Experiments

The synthetic experiments test conventional metric sensitivity and a search implementation; they do not test the full proposed evaluator. The trajectory study is generated by `paper/run_sensitivity.py`; the search check uses `paper/run_experiments.py`. The following real-video measurements and VLM controls use separate retained scripts and manifests.

4.1 Trajectory-defect study

The sensitivity study generates 24 non-collinear paths with 80 frames. For each path, we separately inject translation drift, translation jitter, a local jump, and rotation jitter at five amplitudes, including zero. Noise realizations are shared across amplitudes within each path and family. Only increasing amplitude within a family is ordered; there is no assumed perceptual ordering between families.

We retain ATE, translation and rotation RPE, translation and rotation error percentiles, and derivatives of aligned positional residuals. Derivatives of residuals avoid penalizing genuine acceleration in the reference path. The experiment retains all 480 measurement rows and resamples whole paths for confidence intervals. It measures amplitude sensitivity, not perceptual accuracy.

4.2 Design-search study

The search space contains a log-scaled continuous confidence threshold, an integer temporal window from 2 to 16, and a categorical precision choice. A smooth hidden function contains a narrow threshold optimum, a window optimum, an interaction term, and a small precision effect. Its maximum is computed on a dense grid. The optimizer sees only evaluated points.

Each of 60 trials starts from four shared random observations. Bayesian and random search then receive the same total budgets from 5 through 20. The primary measure is simple regret, the gap between the best value found and the grid maximum. We also report the fraction of trials within 0.05 of that maximum at budget 20.

5 Results

5.1 Reference-free DAVIS development runs

The pilot uses DAGE [9] and Depth Anything 3 (DA3) [8] as sources of camera estimates, not as the proposed evaluation method. We staged 90 DAVIS train/validation sequences with 6,208 frames, preserving split lists and frame hashes. Camera-estimator development uses the first 60 frames of the first three listed training sequences: *bear*, *bmx-bumps*, and *boat*. The separate correspondence study below uses annotated DAVIS validation videos, not camera-estimator results on them. DAGE produces camera poses and normalized intrinsics. On bear, a repeat run retaining calibration reproduced the original poses elementwise. Its runtime was 72.08 seconds including loading, with 7.04 GB peak allocated GPU memory. These development settings do not form a controlled runtime benchmark.

The DA3 runs use query-chunked attention with the same keys and values. Unit tests compare the attention kernel with unchunked evaluation; end-to-end numerical equivalence has not been measured. Native processed-image intrinsics are converted back to source-image coordinates using the audited resize convention. Both methods use the same source frames and decoded inputs.

On bear, mutual SIFT ratio matching retains 22,472 matches over 28 frame pairs at supplied-frame lags of 1, 10, and 30. Sampson residuals use predicted, not independently measured, calibration. Median pair-median residuals are 0.000204, 0.000777, and 0.001353 for the three lags, in normalized camera coordinates. BMX and boat retain 11,626 and 18,258 matches respectively. Cross-method diagnostics instead use symmetric point-to-epipolar-line distance in source pixels to avoid directly comparing residuals in differently scaled normalized camera coordinates. We do not interpret these values as camera accuracy or usability.

We independently change each pair's target-camera rotation by $-15, -5, -1, 0, 1, 5, 15$ degrees around each local axis, holding matches and calibration fixed. The 588 retained measurements are pairwise interventions, not modified full trajectories. A positive 15-degree change increases median residuals on all 28 pairs for each axis.

This demonstrates response to the selected interventions, not superiority to existing metrics or agreement with people.

5.2 Ambiguity and placement-evidence coverage

Reversing all predicted camera centers changes the essential matrix only by sign. Across all 168 native pairs from six outputs, absolute epipolar residuals therefore remain unchanged. A conventional positive-depth check distinguishes that sign ambiguity on nondegenerate static correspondences. It is not our contribution and cannot by itself certify an estimate: predicted calibration, mismatches, small parallax, and moving surfaces still affect the result.

Table 1 reports early-fit placement measurements with their available support. The same image observations are used by both camera methods within each row. Across rows, changing the image tracker changes the subset; a lower or higher median is not evidence that the evaluator improved. Median errors also omit unavailable fits, whose counts must remain visible. On BMX, LK supplies only three common later observations and all 12 tracks are lost by the final frame. CoTracker supplies 98 common observations, but this is still only 16.7% of the planned evidence. On boat, the CoTracker arm has ten valid early fits for DAGE and six for DA3; the remaining two and six fits respectively have nonpositive depth. These are unresolved diagnostic cases, not established functional failures.

Offline tracking changed early coordinates when future images were supplied. Maximum differences between prefix-only and full-video CoTracker coordinates over the fitting frames were 1.642 pixels on bear, 12.923 on BMX, and 8.483 on boat. These are differences between predictions, not measured tracking errors. Separate prefix inference avoids using the full-video early coordinates in the fit; it does not make the later tracks ground truth.

5.3 Validating correspondence evidence

Point tracks should be checked independently before they are used to judge camera estimates. We use TAP-Vid DAVIS [3], whose point-position and visibility annotations were produced with human review and optical-flow assistance. These annotations do not establish staticness or camera accuracy. Before evaluating an image check, we partition the 30 videos by a deterministic hash of the video name: ten development videos and twenty reserved videos. This is a project-specific split, not a result on the standard full TAP-Vid benchmark.

Each annotated track is initialized once at its first visible point. LK then runs independently at native image resolution, without later annotation inputs or restart after loss. We compare four fixed admission rules: any available estimate; forward–backward error at most one native pixel; normalized cross-correlation (NCC) of at least 0.8 with the starting grayscale patch; and both checks. NCC uses a 17×17 subpixel patch without scale or rotation alignment. Image-border and textureless patches cannot pass the patch check. The rules are fixed before scoring the reserved videos.

For scoring only, an accepted estimate is correct if the annotated point is visible and its Euclidean error is below four pixels after scaling coordinates to 256×256. Accepted occluded estimates are errors. Precision is the correct fraction among accepted estimates; visible-correct recall is the correctly accepted fraction of all visible target observations. We compute each quantity per video and then average across videos. Rejected and unavailable observations remain in the recall denominator. This scoring tolerance must not be confused with a four-native-pixel geometric bound.

The reserved set contains 435 first-visible queries and 26,374 subsequent observations. Table 2 reports all four rules. The joint rule achieves 96.2% mean precision and 24.0% mean visible-correct recall. Descriptive 95% percentile intervals from 20,000 video-bootstrap resamples are 94.3–97.8% and 16.6–33.0%, respectively. Every reserved video has at least one accepted estimate. Pooled counts, which are not the macro averages, comprise 4,245 accepted estimates: 4,133 correct, 97 visible but outside tolerance, and 15 occluded. The source-patch check reduces incorrect admission while discarding substantial evidence. It is a classical baseline, not the claimed novel evaluator.

This experiment validates a correspondence-admission component under the stated tolerance, not a camera-failure rule. Image agreement does not prove that a surface is static, and occasional accepted errors can invalidate a multi-frame geometric constraint. Annotation uncertainty, native-pixel tolerances, and simultaneous whole-track reliability must be addressed before interpreting geometric inconsistency as camera failure. The model-training study remains outside these reported results.

5.4 Conventional metric sensitivity checks

ATE and translation RPE increase with amplitude in all tested translation-corruption comparisons. For rotation-only corruption, ATE remains tied, while rotation RPE increases in all tested comparisons. These are implementation sensitivity checks. Standard RPE already detects the selected rotation defect, so this result does not demonstrate an advantage over a conventional multi-metric baseline.

Rotation jitter preserves camera centers by construction. Its zero ATE is therefore expected. The experiment checks that position and rotation measurements respond as intended.

Whether persistent jitter is worse than a brief jump depends on the task and severity. Independent judgments

Table 1: Development placement diagnostics. Each row uses common available background observations for the two camera estimates. Each clip plans 588 later observations across 12 query slots. Image tracks and annotated background are not ground truth; these values do not establish method rankings.

Clip	Image tracker	Observations / 588	Anchors / 12	DAGE median (px)	DA3 median (px)
bear	LK	494	11	4.936	2.323
bear	CoTracker3	472	11	5.104	2.398
bmx-bumps	LK	3	1	10.516	17.747
bmx-bumps	CoTracker3	98	3	25.973	39.806
boat	LK	127	3	14.212	10.609
boat	CoTracker3	174	4	17.795	12.472

Table 2: Correspondence admission on twenty reserved TAP-Vid DAVIS videos. Percentages are video-macro averages. Recall includes all visible target observations, including those without an estimate.

Rule	Precision (%)	Recall (%)
LK available	49.3	53.0
Forward–backward	53.3	50.9
Starting-patch NCC	95.2	24.7
Both	96.2	24.0

and application outcomes are needed to evaluate that tradeoff. The present results provide no evidence for a universal order.

5.5 Synthetic search implementation check

At budget 20, median regret relative to the dense-grid maximum is 0.059 for Bayesian search and 0.236 for random search. Bayesian search reaches within 0.05 of that maximum in 48.3% of trials; random search does so in 10.0%. This is one synthetic objective, unrelated to measured tracking quality, and is only an optimizer implementation check.

This experiment verifies the implementation on a mixed search space with a known optimum. It does not establish the same gain on a real tracking model. Real objectives may be noisier, less smooth, and more expensive, and preference uncertainty should then be passed into the search model.

6 Reproduction Harness as Supporting Evidence

The evaluator is implemented inside a broader protocol-first reproduction harness. Its canonical ledger currently contains 47 verified published-result cells across 16 method variants and 13 datasets: 39 depth cells and 8 pose cells. The median absolute deviation from the paper value is 1.5%, the 90th percentile is 4.8%, and 46 of 47 cells are within 5%.

These numbers show that the infrastructure can preserve varied model, dataset, alignment, and metric protocols. They are not the main contribution and not a reproduction success rate. The ledger is curated, not randomly sampled, and excludes claims that have not reached a verified state.

7 Required Validation and Limitations

The primary missing experiment is human validation of functional usability on real method outputs. Reviewers must see the same source evidence without method names or metric-derived suggested answers. Pairwise preference, structural validity, and absolute usability are separate labels. Reviewers must be able to abstain and cite an interval or visible failure. Multiple independent reviewers and repeated hidden examples are needed to measure disagreement; project-author review alone is insufficient for the final study.

The final comparison must include ORE where its static-track and calibration requirements are met, simpler geometric checks, conventional pose metrics where references exist, and untuned and tuned judge variants. The primary failure-detection measures are false DNF among human-usable outputs, missed failure among human-unusable outputs, and coverage including all abstentions. Thresholds must be selected on development data and fixed before held-out evaluation. Sequence-level resampling must retain correlated frame pairs together. A claim of cross-method generalization also requires holding out tracker identities, not merely frames from familiar videos.

DAVIS masks do not establish that an object is static, and our staged data does not provide measured intrinsics or camera poses. The current three-clip, two-method camera study is insufficient to evaluate generalization, ranking, or semantic interpretation. The separate correspondence study provides independent position/visibility references but not camera-usability labels. The available author review does not replace independent validation, and no independent reviewers are currently available. This report includes no validated fine-tuning or reinforcement-learning result for camera evaluation.

The trajectory study measures response to injected amplitude. It cannot show that the proposed method matches human preference or improves selection over existing metrics. The highest-priority next experiment is a study using real tracking outputs, independent reviewers, repeated hidden pairs, strong multi-metric baselines, and a held-out judge-calibration set.

The existing pairwise diagnostic uses ground-truth poses; it must not be confused with the reference-free arm. The fixed rubric may not represent every use case. Virtual production, robot navigation, and 3D reconstruction place different costs on short jumps, slow drift, and rotation error. Reporting component judgments is safer than presenting one universal score.

The Bayesian experiment uses a smooth synthetic objective. Gaussian processes can struggle in high-dimensional or conditional spaces. Search can also overfit the evaluator, which is why a held-out final set is required. Finally, the current implementation has not measured inter-rater reliability, learning effects, or vision-language judge calibration on a real comparison set.

8 Conclusion

We describe a reference-free camera-evaluation protocol that separates structural validity, functional usability, and unresolved evidence. Its geometric diagnostics retain observation coverage and distinguish image-track support from camera accuracy. The current experiments characterize these diagnostics but do not yet establish a validated camera-quality metric. The central remaining experiment is whether deterministic measurements combined with separately validated scene interpretation identify functional failures on held-out method outputs. Evaluator training and design search depend on that validation.

Data and Code Availability

The development repository is https://github.com/kmatzen/plumbline. Current local experiments retain scripts, hashes, and raw result records; synchronization of the latest artifacts and a citable release remain pending. Source-containing diagnostic images are retained locally rather than redistributed in the repository. Public model and dataset artifacts remain subject to their original licenses.

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